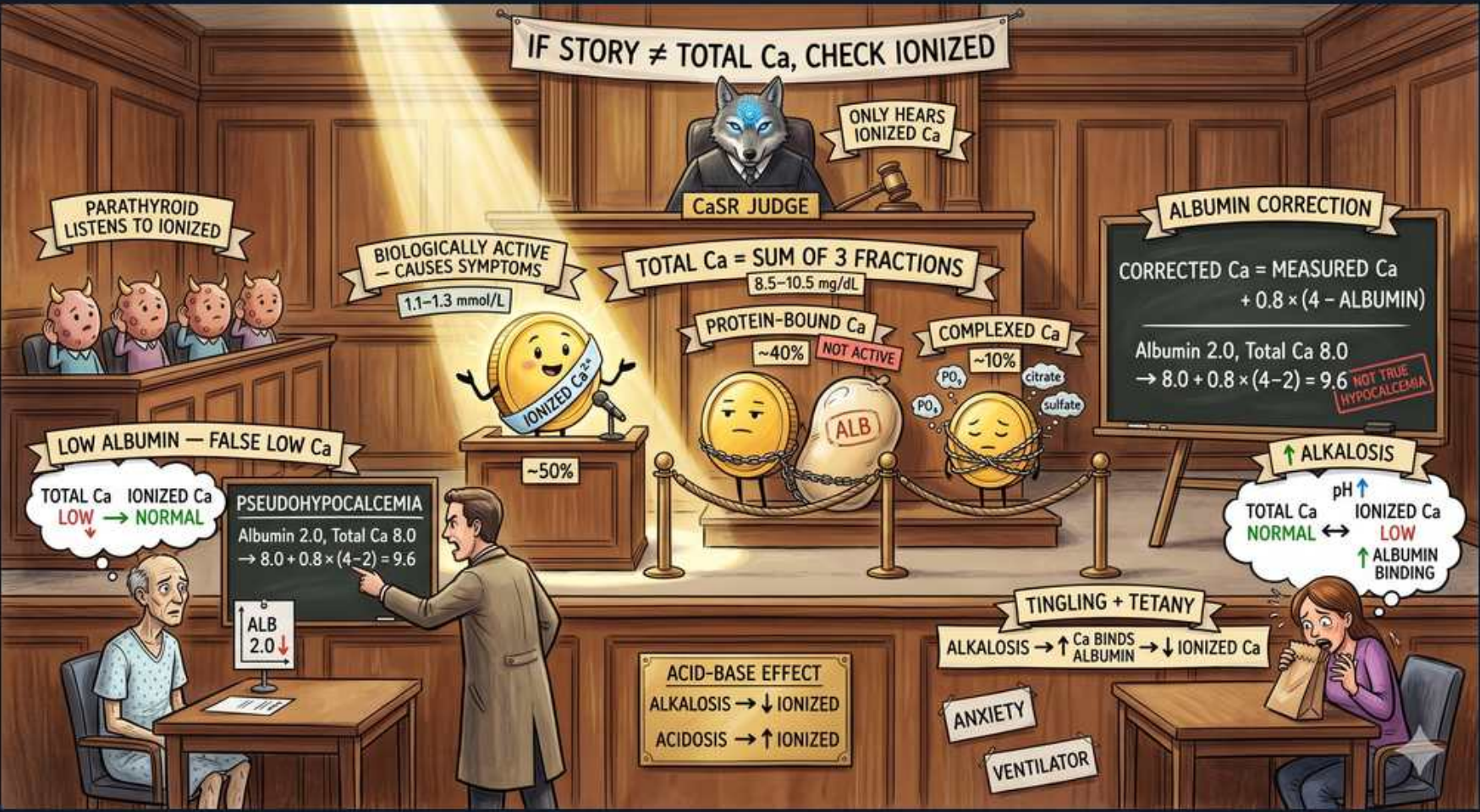


Calcium/PTH/VitD — The Calcium Forms Courtroom



1. If story ≠ total Ca, check ionized (banner)

WHAT YOU SEE

Top banner: 'IF STORY ≠ TOTAL Ca, CHECK IONIZED'.

WHAT IT ENCODES

The unifying rule of the scene: when the clinical picture conflicts with the reported total calcium, measure the IONIZED (free) calcium, which is the physiologically active fraction the body actually senses. Total calcium is confounded by albumin (binding protein) and pH (binding affinity); ionized calcium bypasses both confounders and reflects the true biologic level.

BOARD TRAP

A hyperventilating patient with perioral tingling and a NORMAL total calcium is told 'calcium is normal, this is purely anxiety' — WRONG. Alkalosis can lower ionized calcium with a normal total; the story (tetany) conflicts with the total, so the answer is to check ionized Ca. The discriminator: symptoms-vs-total mismatch → measure ionized.

2. CaSR judge hears only ionized Ca

WHAT YOU SEE

A wolf judge labeled 'CaSR JUDGE — ONLY HEARS IONIZED Ca'.

WHAT IT ENCODES

The calcium-sensing receptor (CaSR), a G-protein-coupled receptor on parathyroid chief cells and renal tubules, senses only IONIZED calcium, not total. When ionized Ca falls, the CaSR signals increased PTH secretion; when it rises, PTH is suppressed. Because the body's entire calcium homeostasis is driven by ionized levels, that fraction is what matters clinically.

BOARD TRAP

A case reasons that a low TOTAL calcium (from hypoalbuminemia) should drive PTH up — WRONG. The CaSR responds only to ionized calcium, which is normal in hypoalbuminemia, so PTH stays normal. The discriminator: PTH tracks ionized, not total; pseudohypocalcemia does not stimulate PTH.

3. Parathyroid listens to ionized

WHAT YOU SEE

Left box: 'PARATHYROID LISTENS TO IONIZED'.

WHAT IT ENCODES

PTH secretion is governed by ionized calcium via the parathyroid CaSR: a drop in ionized Ca within minutes triggers PTH release, which raises calcium by mobilizing bone, increasing renal calcium reabsorption, and activating renal 1-alpha-hydroxylase (→ more calcitriol → gut absorption). PTH therefore tracks the free fraction, explaining why ionized changes (e.g., from alkalosis) drive symptoms even when total Ca is unchanged.

BOARD TRAP

A patient with acute respiratory alkalosis and tetany is expected to have a HIGH PTH because 'calcium is low' — possibly WRONG/misleading. Total calcium is normal so the lab may look unremarkable, and the acute ionized drop may not yet have raised measured PTH dramatically. The discriminator: it is ionized Ca that the parathyroid senses; interpret PTH against ionized, not total.

4. Biologically active = causes symptoms

WHAT YOU SEE

Box: 'BIOLOGICALLY ACTIVE — CAUSES SYMPTOMS, 1.1-1.3 mmol/L'.

WHAT IT ENCODES

Ionized (free) calcium is the ONLY biologically active form, responsible for neuromuscular excitability, cardiac conduction, clotting, and hormone secretion. Normal ionized range is ~1.1-1.3 mmol/L (about 4.5-5.3 mg/dL). Symptoms — tetany, paresthesias, arrhythmia — track ionized calcium, which is why a normal total with a low ionized still produces full-blown symptoms.

BOARD TRAP

A symptomatic patient with a normal total calcium is reassured no treatment is needed — WRONG if ionized is low. Symptoms reflect the ionized fraction, not the total. The discriminator: in a symptomatic patient, the ionized level (not the total) determines whether they are truly hypo/hypercalcemic and need treatment.

5. Total Ca = sum of 3 fractions

WHAT YOU SEE

Center coin: 'TOTAL Ca = SUM OF 3 FRACTIONS, 8.5-10.5 mg/dL'.

WHAT IT ENCODES

Total serum calcium (normal ~8.5-10.5 mg/dL) is the SUM of three fractions: ionized/free (~50%, active), protein-bound (~40%, mostly to albumin, inactive), and complexed to anions (~10%, to phosphate/citrate/bicarbonate, inactive). Standard chemistry panels measure TOTAL, which is why albumin and pH must be considered to infer the active ionized portion.

BOARD TRAP

A reported total calcium is treated as if it equals the active calcium — WRONG when albumin or pH is abnormal. Only ~50% of total is the active ionized fraction, and that proportion shifts with albumin and pH. The discriminator: total is a composite; do not equate it with biologic activity without correcting for albumin (or measuring ionized).

6. Ionized ~50%

WHAT YOU SEE

A glowing coin labeled '~50% IONIZED'.

WHAT IT ENCODES

About 50% of total calcium circulates as free ionized Ca⁺⁺, the glowing 'active' coin. This is the fraction the CaSR senses and the fraction that produces clinical effects. It is the gold-standard measurement when total calcium is unreliable (abnormal albumin, acid-base disturbance, citrate load, critical illness).

BOARD TRAP

An ICU patient's total calcium is used to guide replacement during massive transfusion — WRONG. Citrate from blood products chelates ionized calcium, dropping the active ~50% fraction while total may appear normal. The discriminator: in transfusion/critical illness, trust the directly measured ionized calcium, not the total.

7. Protein-bound ~40% (albumin)

WHAT YOU SEE

Coin: 'PROTEIN-BOUND ~40%, NOT ACTIVE, ALB'.

WHAT IT ENCODES

About 40% of calcium is bound to plasma proteins, predominantly albumin (a small amount to globulins), and is biologically INACTIVE. Because this pool is large, changes in albumin shift TOTAL calcium substantially without changing the active ionized level — the mechanism behind pseudohypo/hypercalcemia. Each 1 g/dL fall in albumin lowers total Ca by ~0.8 mg/dL.

BOARD TRAP

A malnourished/cirrhotic patient with low albumin and low total calcium is treated with IV calcium for 'hypocalcemia' — WRONG. The low total is from low albumin (less binding protein); ionized is normal, so this is pseudohypocalcemia needing no treatment. The discriminator: correct for albumin (or measure ionized) before treating a low total.

8. Complexed ~10% (PO4/citrate)

WHAT YOU SEE

Coin: 'COMPLEXED ~10%, PO4, citrate'.

WHAT IT ENCODES

About 10% of calcium is complexed (chelated) to small anions — phosphate, citrate, bicarbonate, lactate — and is inactive. This pool explains how citrate loads or hyperphosphatemia can lower ionized calcium: anions grab free Ca^{++} . It also underlies citrate anticoagulation in apheresis/CRRT and the hypocalcemia of massive transfusion.

BOARD TRAP

A patient receiving massive transfusion or on regional citrate CRRT develops tetany, but their TOTAL calcium is normal so hypocalcemia is excluded — WRONG. Citrate complexes (chelates) ionized calcium, causing symptomatic hypocalcemia with a normal-appearing total. The discriminator: with citrate load, follow ionized calcium and replete it, ignoring the falsely reassuring total.

9. Low albumin = false low Ca

WHAT YOU SEE

Box: 'LOW ALBUMIN — FALSE LOW Ca' over an elderly woman.

WHAT IT ENCODES

Hypoalbuminemia (malnutrition, cirrhosis, nephrotic syndrome, critical illness) lowers TOTAL calcium because there is less albumin to bind the ~40% protein-bound pool, but the active IONIZED fraction stays normal. This is pseudohypocalcemia — a lab artifact, not a true calcium deficit, requiring no calcium treatment.

BOARD TRAP

A hospitalized patient with albumin 2.5 and total Ca 7.8 is diagnosed with true hypocalcemia and given calcium — WRONG. Corrected Ca = $7.8 + 0.8 \times (4 - 2.5) = 9.0$ (normal); this is pseudohypocalcemia. The discriminator: always albumin-correct (or measure ionized) before calling a low total calcium 'real.'

10. Albumin correction formula

WHAT YOU SEE

Chalkboard: 'CORRECTED Ca = MEASURED Ca + $0.8 \times (4 - \text{ALBUMIN})$ ', example 9.6.

WHAT IT ENCODES

Corrected calcium = measured total Ca + $0.8 \times (4.0 - \text{serum albumin in g/dL})$. It estimates what the total calcium would be if albumin were normal (4 g/dL), unmasking pseudohypo/hypercalcemia. Example on the board: albumin 2.0, total Ca 8.0 → $8.0 + 0.8 \times (4 - 2) = 9.6$ mg/dL (normal). The formula assumes a normal albumin of 4.0.

BOARD TRAP

A clinician trusts the correction formula in a critically ill or acid-base-disturbed patient and forgoes the ionized level — WRONG. The formula is only an estimate, unreliable in critical illness, abnormal pH, paraproteinemia, and citrate load. The discriminator: when accuracy matters or the formula and clinical picture disagree, measure ionized calcium directly.

11. Pseudohypocalcemia worked example

WHAT YOU SEE

Box: 'PSEUDOHYPOCALCEMIA — albumin 2.0, total Ca 8.0 → 9.6'.

WHAT IT ENCODES

Worked example: albumin 2.0 g/dL with total Ca 8.0 mg/dL corrects to $8.0 + 0.8 \times (4 - 2) = 9.6$ mg/dL, which is normal. This is pseudohypocalcemia: the low total is entirely explained by the low binding protein, ionized is normal, and NO treatment is required. Reserve calcium therapy for true (corrected/ionized) hypocalcemia.

BOARD TRAP

This patient (corrected Ca 9.6) is still given IV calcium gluconate because the uncorrected value 'looked low' — WRONG and potentially harmful. After albumin correction the calcium is normal. The discriminator: do the correction math first; treating pseudohypocalcemia risks iatrogenic hypercalcemia.

12. Alkalosis → ↑ binding → ↓ ionized

WHAT YOU SEE

Right box: 'ALKALOSIS — total Ca NORMAL, pH up, ionized LOW, albumin binding'.

WHAT IT ENCODES

Alkalosis (high pH) deprotonates albumin's binding sites, INCREASING calcium binding to albumin and thereby LOWERING the free ionized fraction — while TOTAL calcium stays the same. This shifts active calcium into the inactive bound pool, producing symptoms (tetany) despite a normal total. The classic acute trigger is hyperventilation-induced respiratory alkalosis.

BOARD TRAP

A patient with respiratory alkalosis and tetany has a normal total calcium, so a calcium problem is excluded and only sedation is given — WRONG. Alkalosis lowers ionized calcium with a normal total. The discriminator: in alkalosis-driven tetany, measure/treat ionized calcium and correct the alkalosis; do not be reassured by the normal total.

13. Acid-base effect summary

WHAT YOU SEE

Box: 'ACID-BASE EFFECT — alkalosis ↑ ionized binding (↓ free), acidosis ↑ ionized free'.

WHAT IT ENCODES

pH inversely controls the ionized fraction by competing with calcium for albumin's binding sites. ALKALOSIS (fewer H^+ on albumin) increases Ca-albumin binding → LOWERS ionized (→ tetany). ACIDOSIS (more H^+ displacing Ca) decreases binding → RAISES ionized free calcium. Total calcium is unchanged in both; only the active fraction shifts.

BOARD TRAP

When correcting a patient's chronic acidosis (e.g., rapid bicarbonate in DKA or renal failure), no thought is given to calcium and the patient develops tetany — WRONG to ignore. Rapidly raising pH increases Ca-albumin binding and drops ionized calcium acutely. The discriminator: alkalosis lowers, acidosis raises ionized Ca — anticipate ionized changes when you shift pH.

14. Tingling/tetany with alkalosis

WHAT YOU SEE

Box: 'TINGLING + TETANY — alkalosis ↑ Ca-albumin binding → ↓ ionized', anxious patient hyperventilating.

WHAT IT ENCODES

Hyperventilation (anxiety, panic) causes respiratory alkalosis, which increases calcium binding to albumin and lowers ionized calcium, producing perioral and acral tingling, carpopedal spasm, and frank tetany — all with a NORMAL total calcium. Treatment is to correct the hyperventilation (reassurance, rebreathing) rather than to infuse calcium.

BOARD TRAP

A panic-attack patient with perioral tingling and carpopedal spasm is loaded with IV calcium for presumed true hypocalcemia — WRONG. The tetany is from an acute ionized drop due to respiratory alkalosis, with a normal total and normal total-body calcium. The discriminator: treat the alkalosis/hyperventilation; calcium stores are normal, so IV calcium is unnecessary.
